

Set 1 - Loops (for, while, do-while, nested, break-continue)

Q1.

Write a Java program using a for loop to display the daily step count for each day of a week, where the step counts are stored in an array.

Answer:

```
public class StepCounter {
    public static void main(String[] args) {
        int[] steps = {5230, 7840, 6100, 9520, 4300, 11200, 8000};
        String[] days = {"Monday", "Tuesday", "Wednesday", "Thursday",
                        "Friday", "Saturday", "Sunday"};

        for (int i = 0; i < steps.length; i++) {
            System.out.println(days[i] + ": " + steps[i] + " steps");
        }
    }
}

/* Sample Output:
Monday: 5230 steps
Tuesday: 7840 steps
Wednesday: 6100 steps
Thursday: 9520 steps
Friday: 4300 steps
Saturday: 11200 steps
Sunday: 8000 steps
*/
```

Q2.

Simulate a Zomato-style order summary: Given an array of order amounts for a month, use a while loop to calculate and print the total amount spent.

Answer:

```
public class ZomatoOrderSummary {
    public static void main(String[] args) {
        double[] orderAmounts = {350.50, 220.00, 480.75, 199.99, 610.20, 150.00};

        int i = 0;
        double total = 0.0;

        while (i < orderAmounts.length) {
            total += orderAmounts[i];
            i++;
        }

        System.out.println("Total number of orders: " + orderAmounts.length);
        System.out.println("Total amount spent this month: Rs. " + total);
    }
}

/* Sample Output:
```

```
Total number of orders: 6
Total amount spent this month: Rs. 2011.44
*/
```

Q3.

Create a Java program that uses a do-while loop to repeatedly ask the user to enter their favorite music artist until they type 'exit', then print how many artists they entered.

Answer:

```
import java.util.Scanner;

public class FavoriteArtists {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int count = 0;
        String artist;

        do {
            System.out.print("Enter your favorite music artist (type 'exit' to stop): ");
            artist = sc.nextLine();

            if (!artist.equalsIgnoreCase("exit")) {
                count++;
            }
        } while (!artist.equalsIgnoreCase("exit"));

        System.out.println("You entered " + count + " artist(s).");
        sc.close();
    }
}

/* Sample Run:
Enter your favorite music artist (type 'exit' to stop): Arijit Singh
Enter your favorite music artist (type 'exit' to stop): Taylor Swift
Enter your favorite music artist (type 'exit' to stop): exit
You entered 2 artist(s).
*/
```

Q4.

Build a nested loop that prints a table showing the number of likes received on Instagram posts for each week (4 weeks, 7 days per week). Hint: Use an outer loop for weeks and an inner loop for days, and generate random like counts for each cell.

Answer:

```
import java.util.Random;

public class InstagramLikesTable {
    public static void main(String[] args) {
        Random rand = new Random();
        int weeks = 4;
        int daysPerWeek = 7;

        System.out.println("Instagram Likes Table (Weeks vs Days)");
        System.out.println("-----");

        for (int week = 1; week <= weeks; week++) {
            System.out.print("Week " + week + ": ");
```

```

        for (int day = 1; day <= daysPerWeek; day++) {
            int likes = rand.nextInt(500) + 1; // random likes between 1 and 500
            System.out.print(likes + "\t");
        }
        System.out.println();
    }
}

/* Sample Output:
Instagram Likes Table (Weeks vs Days)
-----
Week 1: 245    102    389    77    412    19    301
Week 2: 88     460    233    150   76     499    12
Week 3: 310    45     120    400   275    66     190
Week 4: 501    ...(values vary each run since they are random)
*/

```

Q5.

Write a Java program that iterates through an array of cricket scores (one per match) and skips any score below 50 using continue, but stops processing completely if a score of 200 is found using break.

Answer:

```

public class CricketScores {
    public static void main(String[] args) {
        int[] scores = {45, 78, 120, 30, 200, 65, 90};

        for (int i = 0; i < scores.length; i++) {
            if (scores[i] < 50) {
                continue; // skip low scores
            }
            if (scores[i] == 200) {
                System.out.println("Score of 200 found! Stopping processing.");
                break; // stop completely
            }
            System.out.println("Valid score: " + scores[i]);
        }
    }
}

/* Sample Output:
Valid score: 78
Valid score: 120
Score of 200 found! Stopping processing.
*/

```